Likelihood ratio test : $N(\underline{\mu}, \Sigma)$; $H_0 : \underline{\mu} = \underline{\mu}_0$

$$\lambda = \frac{\underset{H_0}{\text{Max}}\, L}{\underset{\Omega}{\text{Max}}\, L} = \frac{\text{Max } L(\underline{\mu}_0, \Sigma)}{\text{Max } L(\underline{\mu}, \Sigma)}$$

Parametric space under $H_0 = \hat{\Omega}_{H_0} = (\underline{\mu}_0, \hat{\Sigma}_{00})$

whole space under Ω = $\hat{\Omega}_{\Omega} = (\hat{\underline{\mu}}_{\Omega}, \hat{\Sigma}_{\Omega})$

$$\hat{\underline{\mu}}_{\Omega} = \bar{\underline{x}} \qquad \hat{\Sigma}_{\Omega} = \frac{A}{N} \qquad \hat{\Sigma}_{00} = \frac{1}{N}\sum_{\alpha=1}^{N}(\underline{x}_\alpha - \underline{\mu}_0)(\underline{x}_\alpha - \underline{\mu}_0)'$$

⇒ Result: For any $|B| \neq 0$

$$\begin{vmatrix} B & C \\ D & E \end{vmatrix} = \begin{vmatrix} B & C \\ D & E \end{vmatrix}\begin{vmatrix} I & -B^{-1}C \\ 0 & I \end{vmatrix} = \begin{vmatrix} B & 0 \\ D & E - DB^{-1}C \end{vmatrix} = |B|\,|E - DB^{-1}C|$$

Eg:
$$\frac{|A|}{|A + \{\sqrt{N}(\bar{\underline{x}} - \underline{\mu})\}\{\sqrt{N}(\bar{\underline{x}} - \underline{\mu})\}'|}$$

$$= \frac{|A|}{\begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu}) \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu}) & I \end{vmatrix}}$$

$$= \frac{|A|}{\begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu}) \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu}) & I \end{vmatrix}\begin{vmatrix} I & -\sqrt{N}A^{-1}(\bar{\underline{x}} - \underline{\mu}) \\ 0 & I \end{vmatrix}}$$

$$= \frac{|A|}{\begin{vmatrix} A & -\sqrt{N}(\bar{\underline{x}} - \underline{\mu}) + \sqrt{N}(\bar{\underline{x}} - \underline{\mu}) \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu}) & N(\bar{\underline{x}} - \underline{\mu})'A^{-1}(\bar{\underline{x}} - \underline{\mu}) + I \end{vmatrix}}$$

Likelihood ratio test :

$\underline{x} \sim N_p(\underline{\mu}, \Sigma)$ $\quad \underline{x}_1, \underline{x}_2, \dots, \underline{x}_N$ sample

$$L = L(\underline{\mu}, \Sigma) = \frac{1}{(2\pi)^{\frac{Np}{2}} |\Sigma|^{\frac{N}{2}}} \exp\left[-\frac{1}{2}\sum_{\alpha=1}^{N}(\underline{x}_\alpha - \underline{\mu})'\Sigma^{-1}(\underline{x}_\alpha - \underline{\mu})\right]$$

LR test statistic $\lambda = \dfrac{\underset{H_0}{\text{Max}}\, L}{\underset{\Omega}{\text{Max}}\, L}$

$\hat{\Omega}_{H_0} = (\hat{\underline{\mu}}, \hat{\Sigma}) = (\underline{\mu}_0, \hat{\Sigma}_\omega)$

$\hat{\Omega}_\Omega = (\hat{\underline{\mu}}_\Omega, \hat{\Sigma}_\Omega) = (\bar{\underline{x}}, \frac{A}{N})$ $\quad \hat{\Sigma}_\Omega = \frac{1}{N}\sum_{\alpha=1}^{N}(\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})'$

$\hat{\Sigma}_\omega = \frac{1}{N}\sum_{\alpha=1}^{N}(\underline{x}_\alpha - \underline{\mu}_0)(\underline{x}_\alpha - \underline{\mu}_0)'$

$$\underset{H_0}{\text{Max}}\, L = \frac{1}{(2\pi)^{\frac{Np}{2}}|\hat{\Sigma}_\omega|^{N/2}} \exp\left[-\frac{1}{2}\sum_{\alpha=1}^{N}(\underline{x}_\alpha - \underline{\mu}_0)'\hat{\Sigma}_\omega^{-1}(\underline{x}_\alpha - \underline{\mu}_0)\right]$$

$$= \frac{1}{(2\pi)^{\frac{Np}{2}}|\hat{\Sigma}_\omega|^{N/2}} \exp\left(-\frac{1}{2}\, tr\left(\hat{\Sigma}_\omega^{-1} \underbrace{\sum_{\alpha=1}^{N}\{(\underline{x}_\alpha - \underline{\mu}_0)(\underline{x}_\alpha - \underline{\mu}_0)'\}}_{N\hat{\Sigma}_\omega}\right)\right)$$

$$= \frac{1}{(2\pi)^{Np/2}|\hat{\Sigma}_\omega|^{N/2}} \exp\left(-\frac{N}{2}\cdot \underbrace{tr\, \hat{\Sigma}_\omega^{-1}\hat{\Sigma}_\omega}_{tr(I_p)}\right)$$

$$= \frac{1}{(2\pi)^{\frac{Np}{2}}|\hat{\Sigma}_\omega|^{N/2}} \exp\left(-\frac{Np}{2}\right)$$

$$A = \sum_{\alpha=1}^{N}(\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})'$$

$$A_0 = \sum_{\alpha=1}^{N}(\underline{x}_\alpha \underset{\pm\bar{\underline{x}}}{-} \underline{\mu}_0)(\underline{x}_\alpha \underset{\pm\bar{\underline{x}}}{-} \underline{\mu}_0)' = A_0 + N(\bar{\underline{x}} - \underline{\mu}_0)(\bar{\underline{x}} - \underline{\mu}_0)'$$

$$\underset{\Omega}{\text{Max}}\, L = \frac{1}{(2\pi)^{\frac{Np}{2}}|\hat{\Sigma}_\Omega|^{\frac{N}{2}}} \exp\left[-\frac{1}{2}\Sigma(\underline{x}_\alpha - \bar{\underline{x}})'\hat{\Sigma}_\Omega^{-1}(\underline{x}_\alpha - \bar{\underline{x}})\right]$$

= (... similar steps as above...)

$$= \frac{1}{(2\pi)^{\frac{Np}{2}}|\hat{\Sigma}_\Omega|^{N/2}} \exp\left(-\frac{Np}{2}\right)$$

$$\lambda = \frac{|\hat{\Sigma}_\Omega|^{N/2}}{|\hat{\Sigma}_\omega|^{N/2}} \quad \text{or:} \quad \lambda^{2/N} = \frac{|\hat{\Sigma}_\Omega|}{|\hat{\Sigma}_\omega|}$$

$$\lambda^{2/N} = \frac{|\Sigma(\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})'|}{|\Sigma(\underline{x}_\alpha - \underline{\mu}_0)(\underline{x}_\alpha - \underline{\mu}_0)'|}$$

$$\lambda^{2/N} = \frac{|A|}{|A + N(\underline{\bar{x}} - \underline{\mu}_0)(\underline{\bar{x}} - \underline{\mu}_0)'|}$$

$$= \frac{|A|}{|A|\left(1 + N(\underline{\bar{x}} - \underline{\mu}_0)' A^{-1} (\underline{\bar{x}} - \underline{\mu}_0)\right)} \quad \downarrow \text{ explained after finishing } (**)$$

$$= \frac{1}{1 + \frac{T^2}{N-1}}$$

where

$$T^2 = \overset{(N-1)}{N} (\underline{\bar{x}} - \underline{\mu}_0)' A^{-1} (\underline{\bar{x}} - \underline{\mu}_0)$$

$$= (N) (\underline{\bar{x}} - \underline{\mu}_0)' S^{-1} (\underline{\bar{x}} - \underline{\mu}_0)$$

$$S = \frac{A}{N-1}$$

Explanation of (**)

$$\frac{|A|}{|A + \{\sqrt{N}(\underline{\bar{x}} - \underline{\mu}_0)\}\{\sqrt{N}(\underline{\bar{x}} - \underline{\mu}_0)\}'|}$$

$$\lambda^{2/N} = \frac{1}{1 + \frac{T^2}{N-1}}$$

Reject H_0 if $\lambda \leq \lambda_0$

λ_0 is chosen such that

$$P(\lambda \leq \lambda_0) = \alpha$$

$$\lambda \leq \lambda_0 \Rightarrow \lambda^{N/2} \leq \lambda_0^{N/2}$$

$$\Rightarrow \lambda^{-2/N} - 1 \geq \lambda_0^{-2/N} - 1$$

$$\Rightarrow (N-1)(\lambda^{-2/N} - 1) \geq (N-1)(\lambda_0^{-2/N} - 1)$$

$$\Rightarrow (N-1)\left(1 + \frac{T^2}{N+1} - 1\right) \geq (N-1)\left(1 + \frac{T_0^2}{N+1} - 1\right)$$

$$\Rightarrow T^2 \geq T_0^2(\alpha)$$

where $T_0^2(\alpha)$ is a constant to be determined by the size condition

$$T_0^2 = (N-1)(\lambda_0^{-2/N} - 1)$$

$$\mathbb{P}(\lambda \leq \lambda_0) = \alpha$$

NOTE: If Σ is known $(\underline{\bar{x}} - \underline{\mu}_0)^T \Sigma^{-1} (\underline{x} - \underline{\mu}_0) \sim \chi^2(p)$

Distribution of T^2

Let $\underset{\sim}{Y} \sim N_p(\underline{\gamma}, \Sigma)$ and $A = nS \sim W(n, \Sigma)$

$\underset{\sim}{Y}$ and S are independent.

If $T^2 = \underset{\sim}{y}^T S^{-1} \underline{y}$ then $\frac{T^2}{n} \cdot \frac{n-p+1}{p} \sim F(p, n-p+1, \delta)$

non-centrality parameter

Pf:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \quad a^2 + b^2 = 1$$

orthogonal matrix

$$c^2 + d^2 = 1$$

$$ac + bd = 0$$

$q_{ij} \quad q_{ji}$

$$\sum_{j \neq i} q_{ij} q_{ji} = 0$$

Let D be a non-singular matrix such that $D \Sigma D' = I$.

$\Rightarrow$ Define $Y^* = DY$, $S^* = DSD'$, $\gamma^* = D\gamma$, $Z^* = DZ$

(Lemma: For any $p \times p$ non-singular matrix C and H and any vector $\underline{k}$, $\underline{k}^T H^{-1} \underline{k} = (C\underline{k})' (CHC')^{-1} (C\underline{k})$

RHS: $\underline{k}' C' C'^{-1} H C^{-1} C \underline{k} = \underline{k}' H^{-1} \underline{k}$)

$$T^2 = \underline{Y}^{*\prime} S^{*-1} \underline{Y}^* = ((D\underline{Y})'(D\Sigma D')^{-1}(D\underline{Y}) = Y^T D^T D^{T-1} S D^{-1} D Y$$

$$= Y^T S Y$$

$\Rightarrow \underset{\sim}{Y}^* \sim N_p(\underline{\gamma}^*, D\Sigma D' = I)$

$\Rightarrow \underset{\sim}{Y}^*$ and nS^* are independently distributed as $\sum_{\alpha=1}^{n} Z_\alpha^* Z_\alpha^{*\prime} = \sum_{\alpha=1}^{n} (DZ)_\alpha^* (DZ)_\alpha'$; Each $Z_\alpha^* \sim N(0, I)$

- Define an orthogonal matrix $Q_{p \times p}$ st. first row $q_{1i} = \frac{Y_i^*}{\sqrt{\underline{Y}^{*\prime} \underline{Y}^*}}$ $i = 1, 2, \ldots, p$

remaining rows can be defined arbitrarily

(Lemma: $A_{n\times m}$ $(n > m)$ s.t. $A'A = I_m$, then $\exists$ a matrix B s.t. $(A \; B)$ is an orthogonal matrix).

$Q \rightarrow$ depends on $\underline{Y}^*$ (random) $\Rightarrow Q$: random matrix

Let $U = Q\underline{Y}^*$, $B = Q(nS)Q'$

$U_1 = \sum_i q_{1i} Y_i^* = \sqrt{\underline{Y}^{*\prime}\underline{Y}^*}$

$U_j = \sum_i q_{ji} Y_i^* = \sqrt{\underline{Y}^{*\prime}\underline{Y}^*} \sum q_{ji} q_{1i} = 0$

$\left(\frac{T^2}{n} = \underline{Y}^{*\prime}(Q^{-1}Q)' S^{-1} (Q^{-1}Q)\underline{Y}^* = \underline{Y}^{*T} S^{-1} \underline{Y}^*\right)$

$$\frac{T^2}{n} = U^T B^{-1} U = (U_1 \; 0 \; 0 \cdots 0)\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ | & | & & | \\ b_{p1} & b_{p2} & \cdots & b_{pp} \end{pmatrix}\begin{pmatrix} U_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$= U_1^2 b_{11} \qquad B^{-1} = ((b_{ij}))$$

$$B = \begin{pmatrix} b_{11} & \underline{b}'_{(1)} \\ \underline{b}_{(1)} & B_{22} \end{pmatrix} \Rightarrow \frac{1}{b_{11}} = b_{11} - \underline{b}'_{(1)} B_{22}^{-1} \underline{b}_{(1)}$$

$$= b_{11.2,3\ldots p}$$

→ Recall: $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = nS$

$$A_{11.2} = A_{11} - A_{12} A_{22}^{-1} A_{21} \sim W_p(n-p+q, \Sigma_{11.2})$$

$b_{11.2,3,\ldots p} \sim \chi^2(n-p+1)$

$\underline{Y}^{*T}\underline{Y}^* \sim$ non-central χ^2

$$\frac{T^2}{n} = \frac{U_1^2}{(1/b_{11})} = \frac{\underline{Y}^{*T}\underline{Y}^*}{b_{11.23\ldots p}} = \frac{\text{non-central } \chi^2}{\chi^2(n-p+1)}$$

$$\frac{T^2}{n} = \frac{p}{n-p+1} \cdot \frac{\text{non-central } \chi^2/p}{\chi^2_{(n-p+1)}/(n-p+1)} \sim \overset{\text{non-central}}{F(p, n-p+1, \gamma)}$$

$$\frac{(n-p+1)}{p}\frac{T^2}{n} \sim \text{non-central } F(p, n-p+1, \gamma' \Sigma^{-1} \gamma)$$

$$\left(\frac{N-p}{p}\right)\left(\frac{T^2}{N-1}\right) \sim \text{non-central } F(p, N-p, \gamma' \Sigma^{-1} \gamma)$$

T^2 distribution with n degrees of freedom.

① <u>One sample mean problem</u>

$X_1, X_2, \ldots, X_N \overset{ind}{\sim} N(\underline{\mu}, \Sigma)$ $\qquad \lambda = \dfrac{1}{1 + \frac{T^2}{N-1}}$

$\lambda \leq \lambda_0 \longrightarrow T^2 \geq T_0^2(\alpha)$

$T_0^2 = (N-1)(\lambda_0^{-2/N} - 1)$

$\underline{Y} = \sqrt{N}(\bar{\underline{X}} - \underline{\mu})$ $\qquad T_0^2 = \dfrac{(N-1)p}{N-p} \cdot F_{p, N-p}(\alpha)$

$= T^2_{p, N-1}(\alpha)$ or $T^2_{p,n}(\alpha)$

→ Confidence region

All $\underline{\mu}$, s.t. $N(\bar{\underline{x}} - \underline{\mu})' S^{-1}(\bar{\underline{x}} - \underline{\mu}) \leq T^2_{p, N-1}(\alpha)$

Lemma: If $B_{p \times p}$ is a non-singular matrix and $\underline{v}$: $p \times 1$ vector the $\underline{v}' B^{-1} \underline{v}$ is the non-zero root of $|\underline{v}\underline{v}' - \lambda B| = 0$

Pf: Let λ_1 is a non-zero root of $|\underline{v}\underline{v}' - \lambda B| = 0$ with characteristic vector $\underline{\beta}$ satisfying $\underline{v}\underline{v}'\underline{\beta} = \lambda B \underline{\beta}$

Since $\lambda_1 \neq 0$, $\underline{v}'\underline{\beta} \neq 0$, we have $\underline{v}' B^{-1} \underline{v}\underline{v}'\underline{\beta}$

$= \underline{v}' B^{-1} \times \lambda B \underline{\beta}$

or $(\underline{v}' B^{-1} \underline{v})(\underline{v}'\underline{\beta}) = \lambda_1(\underline{v}'\underline{\beta})$

Consider $\underline{v} = \sqrt{N}(\bar{\underline{x}} - \underline{\mu}_0)$ $\quad B = A$

T^2 statistics $\rightarrow$ computed from $\bar{\underline{x}}$ and A

$A^{-1}(\bar{\underline{x}} - \underline{\mu}_0) = \underline{b}$ is a solution of $A\underline{b} = \bar{\underline{x}} - \underline{\mu}_0$
$= N(\bar{\underline{x}} - \underline{\mu}_0)\,\underline{b}$

$\dfrac{T^2}{N-1}$ is the non zero root of $|N(\bar{\underline{x}} - \underline{\mu}_0)(\bar{\underline{x}} - \underline{\mu}_0)' - \lambda A| = 0$

② Two sample test

$\underline{x}_1^{(1)}, \underline{x}_2^{(1)}, \ldots \underline{x}_{N_1}^{(1)} \sim N(\underline{\mu}^{(1)}, \Sigma) \rightarrow$ ~~complete~~

$H_0 : \underline{\mu}^{(1)} = \underline{\mu}^{(2)}$

$\underline{x}_1^{(2)}, \underline{x}_2^{(2)}, \ldots \underline{x}_{N_2}^{(2)} \sim N(\underline{\mu}^{(2)}, \Sigma)$

$\Rightarrow$ Compute $\bar{\underline{x}}^{(1)}, S_1$ from first sample

Compute $\bar{\underline{x}}^{(2)}, S_2$ from second sample

Pooled cov. matrix $= \dfrac{(N_1 - 1) S_1 + (N_2 - 1) S_2}{N_1 + N_2 - 2}$

$= \dfrac{n_1 S_1 + n_2 S_2}{n_1 + n_2} = S$

$S_1 = \dfrac{1}{N_1 - 1} \sum_{\alpha=1}^{N_1} (\underline{x}_\alpha^{(1)} - \bar{\underline{x}}^{(1)})(\;)'$ $\qquad S_2 = \dfrac{1}{N_2 - 1} \sum_{\alpha=1}^{N_2} (\underline{x}_\alpha^{(2)} - \bar{\underline{x}}^{(2)})(\;)'$

$V(\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)}) = \dfrac{\Sigma}{N_1} + \dfrac{\Sigma}{N_2} = \dfrac{N_1 + N_2}{N_1 N_2} \Sigma$

$V\left(\underbrace{\sqrt{\dfrac{N_1 N_2}{N_1 + N_2}} (\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)})}_{\underline{y}}\right) = \Sigma$

$\Rightarrow$ Under H_0

$\sqrt{\dfrac{N_1 N_2}{N_1 + N_2}} (\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)}) \sim N(\underline{0}, \Sigma)$

$(N_1 + N_2 - 2) S \sim \sum_{\alpha=1}^{N_1 + N_2 - 2} \underline{Z}_\alpha \underline{Z}_\alpha'$ where $\underline{Z}_\alpha \sim N(0, \Sigma)$

$$T^2 = \frac{N_1 N_2}{N_1+N_2}(\bar{x}^{(1)}-\bar{x}^{(2)})' S^{-1} (\bar{x}^{(1)}-\bar{x}^{(2)}) \sim T^2(N_1+N_2-2)$$

under H_0

→ Under H_0

$$\frac{N_1+N_2-p-1}{p} \cdot \frac{T^2}{N_1+N_2-2} \sim F(p, N_1+N_2-p-1)(\alpha)$$

→ Confidence region : Set of vectors m st.

$$(\bar{x}^{(1)}-\bar{x}^{(2)}-m)' S^{-1} () \leq \frac{N_1+N_2}{N_1 N_2} \cdot T^2_{p, N_1+N_2-2}(\alpha)$$

③ Problem of symmetry (Invariance property of T^2)

$$H_0: \overbrace{\mu_1 = \mu_2 = \dots = \mu_p}^{p \text{ parameters}} \qquad x_1, x_2, \dots x_N \sim N(\mu, \Sigma) \quad \mu = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}$$

$\Updownarrow$

$$H_0: \underbrace{\mu_1-\mu_2=0, \mu_2-\mu_3=0 \dots, \mu_{p-1}-\mu_p=0}_{(p-1) \text{ linearly ind. contrast}}$$

$\alpha_1\mu_1+\alpha_2\mu_2=0$

$\alpha_1+\alpha_2=0$

contrast

$$H_0: \quad C\underline{\epsilon}=0 \qquad C: (p-1)\times p, \quad \underline{\epsilon}=(1,\dots,1)'$$

$$y_\alpha = Cx_\alpha \sim N(C\mu, C\Sigma C')$$

$$H_0: C\mu = 0$$

$$H_0: \mu = 0$$

$$T_x^2 = N(\bar{x}-0)' S^{-1}(\bar{x}-0) = N\bar{x}' S^{-1}\bar{x}$$

$$T_y^2 = N(C\bar{x})'(CSC)^{-1}(C\bar{x}) = N\bar{x}C'C'^{-1}S^{-1}C^{-1}C\bar{x}$$

$$= N\bar{x}'S^{-1}\bar{x}$$

$$= T_x^2$$

T^2 remains invariant under a non-singular liniar transformation.

MAHALANOBI'S DISTANCE

Define : $\Delta^2 = (\underline{\mu}_1 - \underline{\mu}_2)' \Sigma^{-1} (\underline{\mu}_1 - \underline{\mu}_2)$

Distance between two population $N(\underline{\mu}_1, \Sigma)$ & $N(\underline{\mu}_2, \Sigma)$

→ For two vectors $\underline{x} = (x_1, \ldots, x_p)'$ $\mathbb{R}^p$

$\underline{y} = (y_1, \ldots, y_p)'$ $\quad d_s(\underline{x}, \underline{y}) = \sqrt{(\underline{x}-\underline{y})' S^{-1} (\underline{x}-\underline{y})}$

i) $d_s(\underline{x}, \underline{y}) = d_s(\underline{y}, \underline{x})$

ii) $d_s(\underline{x}, \underline{y}) > 0$ if $\underline{x} \neq \underline{y}$

iii) $d_s(\underline{x}, \underline{y}) = 0$ if $\underline{x} = \underline{y}$

iv) $d_s(\underline{x}, \underline{y}) \leq d_s(\underline{x}, \underline{z}) + d(\underline{y}, \underline{z})$ Triangle inequality.

→ Rao's U-statistic

$\underline{y} \sim N(\underline{\mu}, \Sigma)$, $A \sim W(n, \Sigma)$ $\quad \beta = \Sigma_{21} \Sigma_{11}^{-1}$

dist of $\underline{y}$ and A are ind.

Partition : $\underline{y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix} \begin{matrix} (q) \\ (p-q) \end{matrix}$, $\underline{\mu} = \begin{pmatrix} \mu_1 \\ \underline{\mu}_2 \end{pmatrix} \begin{matrix} (q) \\ (p-q) \end{matrix}$, $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$

Consider $\Delta_p^2 = \underline{\mu}' \Sigma^{-1} \underline{\mu}$ $\quad A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$

$$= (\underline{\mu}_1' \ \underline{\mu}_2') \begin{pmatrix} \Sigma_{11}^{-1} + \beta' \Sigma_{22\cdot 1}^{-1} \beta & -\beta' \Sigma_{22\cdot 1}^{-1} \\ -\Sigma_{22\cdot 1}^{-1} \beta & \Sigma_{22\cdot 1}^{-1} \end{pmatrix} \begin{pmatrix} \mu_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$\vdots$

$$= \underline{\mu}_1' \Sigma_{11}^{-1} \underline{\mu}_1 + (\underline{\mu}_2 - \beta \underline{\mu}_2)' \Sigma_{22\cdot 1}^{-1} (\underline{\mu}_2 - \beta \underline{\mu}_2)$$

$$\Delta_p^2 = \Delta_q^2 + (\mu_2 - \beta \mu_1)' \Sigma_{22\cdot 1}^{-1} (\mu_2 - \beta \mu_1)$$

$$T_p^2 = \underline{y}' A^{-1} \underline{y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}' \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^{-1} \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}$$

$$\frac{T_p^2}{N-1} = \frac{T_q^2}{N-1} + \frac{(\underline{y}_2 - \beta \underline{y}_1)' A_{22\cdot 1}^{-1} (\underline{y}_2 - \beta \underline{y}_1)}{N-1}$$ $\quad \beta = A_{21} A_{11}^{-1}$

$$\frac{T_p^2}{n} = \frac{T_q^2}{n} + \underline{z}' A_{22\cdot 1}^{-1} \underline{z} \qquad \underline{z} = \frac{\underline{y}_2 - \beta \underline{y}_1}{\sqrt{N-1}}$$

Rao defined $U = \dfrac{1 + \frac{T_q^2}{n}}{1 + \frac{T_p^2}{n}}$ $\quad n = N-1$

Rao's U statistic

$$\left(\frac{n-p+1}{p-q}\right)\left(\frac{1}{U} - 1\right) \sim F(p-q,\, n-p+1) \text{ when } \Delta_p^2 = \Delta_q^2$$

Δ_p^2 = distance between 2 population with all p variables

Δ_q^2 = " " 2 " " q variables

$H_0: \Delta_p^2 = \Delta_q^2 \qquad p > q$

(I) First q factors are equally informative

→ $(p-q)$ variables are not useful.

$\Delta_p^2 = \Delta_q^2$ if $\underline{\mu}_2 - \beta \underline{\mu}_1 = 0$ or $\Sigma_{22\cdot 1}^{-1} = 0$

(II) $\underline{\mu}_2 - \beta \underline{\mu}_1 = 0 \quad \therefore \underline{\mu}_2 = \beta \underline{\mu}_1$

i.e. if $\underline{\mu}_1 = \underline{0}$ then $\underline{\mu}_2 = \underline{0}$.

$H_0: \mu_1 = 0$ if $\mu_2 = 0$

(III) the linear combination $\underline{a}'\underline{x}$

$\underline{x}_1 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ st $\underline{x}_0$ and $\underline{x}_1$ are uncorrelated

$Cov(\underline{a}'\underline{x}, x_1) = 0$ or $Cov(\underline{a}_1'\underline{x}_1 + \underline{a}_2'\underline{x}_2, \underline{x}_1) = 0$

$= \underline{a}_1'\Sigma_{11} + \underline{a}_2'\Sigma_{21} = 0$

$\Rightarrow \boxed{\underline{a}_1' = -\underline{a}_2'\beta}$

(iii) $\mathbb{E}(\underline{c}'\underline{x}) = \underline{a}_1'\underline{\mu}_1 + \underline{a}_2'\underline{\mu}_2 = \underline{a}_2'(\underline{\mu}_2' - \beta\underline{\mu}_1)$

if $\mathbb{E}(\underline{a}'\underline{x}) = 0 \ \forall \ \underline{a} \Rightarrow \underline{\mu}_2 - \beta\underline{\mu}_1 = 0$

or $\underline{\mu}_2 = \beta\underline{\mu}_1$

$H_0 : \Delta_p^2 = \Delta_q^2$

Multivariate Behrens Fischer Problem

$H_0 : \underline{\mu}_1 = \underline{\mu}_2 \quad \Rightarrow H_0 : \underline{\mu}_1 - \underline{\mu}_2 = \underline{0}$

2 independent normal distⁿ $N(\underline{\mu}_1, \Sigma_1)$, $N(\underline{\mu}_2, \Sigma_2)$

Σ_1 and Σ_2 - unknown unequal

Sample $\underline{X}_\alpha^{(1)}$; $\alpha = 1, 2, \ldots, N_1$

$\underline{X}_\alpha^{(2)}$; $\alpha = 1, 2, \ldots, N_2$

$$(\bar{\underline{x}}_1 - \bar{\underline{x}}_2) \sim N\left(\underline{\mu}_1 - \underline{\mu}_2, \frac{\Sigma_1}{N_1} + \frac{\Sigma_2}{N_2}\right)$$

Sample covariance matrix

$$\sum_{\alpha=1}^{N_1} (\underline{x}_\alpha^{(1)} - \bar{\underline{x}}_1)(\underline{x}_\alpha^{(1)} - \bar{\underline{x}}_1)' + \sum_{\alpha=1}^{N_2} (\underline{x}_\alpha^{(2)} - \bar{\underline{x}}_2)(\ )'$$

does not have Wishart dist with cov. matrix multiple of $\frac{\Sigma_1}{N_1} + \frac{\Sigma_2}{N_2}$

Define;

$\underline{Y} \sim N(\underline{\mu}^*, \Sigma^*)$ where $\underline{\mu}^* = \underline{\mu}_1 - \underline{\mu}_2$

How do we create $\underline{Y}$?

→ Case 1 : $N_1 = N_2 = N$

$H_0 : \underline{\mu}_1 = \underline{\mu}_2 \qquad H_0 : \underline{\mu}_1 - \underline{\mu}_2 = 0$

$$(\bar{\underline{x}}_1 - \bar{\underline{x}}_2) \sim N\left(\underline{\mu}_1 - \underline{\mu}_2, \frac{\Sigma_1}{N} + \frac{\Sigma_2}{N}\right)$$

Define $\underline{y}_\alpha = \underline{x}_\alpha^{(1)} - \underline{x}_\alpha^{(2)} \sim N(\underbrace{\underline{\mu}_1 - \underline{\mu}_2}_{\underline{\mu}'}, \Sigma_1 + \Sigma_2)$

$\hookrightarrow$ is ind. of $\underline{y}_\beta$ $\beta \neq \alpha$

Let, $\bar{\underline{y}} = \frac{1}{N} \sum_{\alpha=1}^{N} \underline{y}_\alpha$. $S = \frac{1}{N-1} \sum_{\alpha=1}^{N} (\underline{y}_\alpha - \bar{\underline{y}})(\ )'$

$$T^2 = N \bar{\underline{y}}' S^{-1} \bar{\underline{y}} \sim T^2(N-1)$$

There is a loss of $(N-1)$ degrees of freedom.

→ Case 2 : $N_1 \neq N_2$ $(N_1 < N_2)$

$$\underline{y}_\alpha = \left(\underline{x}_\alpha^{(1)} - \sqrt{\frac{N_1}{N_2}} \underline{x}_\alpha^{(2)}\right) + \frac{1}{\sqrt{N_1 N_2}} \left(\sum_{\beta=1}^{N_1} \underline{x}_\beta^{(2)} - \sqrt{\frac{N_1}{N_2}} \sum_{\gamma=1}^{N_2} \underline{x}_\gamma^{(2)}\right)$$

$$= \left(\underline{x}_\alpha^{(1)} - \sqrt{\frac{N_1}{N_2}} \underline{x}_\alpha^{(2)}\right) + \left(\frac{1}{\sqrt{N_1 N_2}} \sum_{\beta=1}^{N_1} \underline{x}_\beta^{(2)} - \frac{1}{N_2} \sum_{\gamma=1}^{N_2} x_\gamma^{(2)}\right)$$

$$\mathbb{E}(\underline{y}_\alpha) = \underline{\mu}_1 - \sqrt{\frac{N_1}{N_2}} \underline{\mu}_2 + \frac{N_1}{\sqrt{N_1 N_2}} \underline{\mu}_2 - \frac{N_1}{N_2} \underline{\mu}_2$$

→ $\mathbb{E}(\underline{y}_\alpha) = (\underline{\mu}_1 - \underline{\mu}_2) = \underline{\mu}'$

→ $\mathrm{Cov}(\underline{y}_\alpha) = \mathbb{E}\left[(\underline{y}_\alpha - \mathbb{E}(\underline{y}_\alpha)) (\underline{y}_\beta - \mathbb{E}(\underline{y}_\beta))'\right]$

$$= \left(\Sigma_1 + \frac{N_1}{N_2} \Sigma_2\right) \delta_{\alpha\beta} \quad , \quad \delta_{\alpha\beta} = \begin{cases} 1 & \text{if } \alpha = \beta \\ 0 & \text{ow} \end{cases}$$

$$= \Sigma^*$$

$H_0 : \mathbb{E}(\underline{y}_\alpha) = \underline{0}$

$$\underset{(N_1-1)\text{ df}}{T^2} = N_1 \bar{\underline{y}}' S^{-1} \bar{\underline{y}} \qquad \bar{\underline{y}} = \frac{1}{N_1} \sum_{\alpha=1}^{N_1} \underline{y}_\alpha = \bar{\underline{x}}_1 - \bar{\underline{x}}_2$$

$$(N_1 - 1) S = \sum_{\alpha=1}^{N_1} (\underline{y}_\alpha - \bar{\underline{y}})(\ )'$$

$H_0 : \sum \beta_i \underline{\mu}_i = \underline{\mu}$

$\beta_1, \beta_2 \ldots \beta_q$: given scalars

$x_\alpha^{(i)} \sim N(\underline{\mu}_i, \Sigma_i) \qquad i = 1, 2, \ldots, q$

$\alpha = 1, 2, \ldots, N_i$

$$\underline{y}_\alpha = \beta_1 \underline{x}_\alpha^{(1)} + \sum_{i=2}^{q} \beta_i \sqrt{\frac{N_1}{N_i}} \left(\underline{x}_\alpha^{(i)} - \frac{1}{N_1} \sum_{\beta=1}^{N_1} \underline{x}_\beta^{(i)} + \frac{1}{\sqrt{N_1 N_i}} \sum_{\gamma=1}^{N_1} x_\gamma^{(i)} \right)$$

$$\mathbb{E}(\underline{y}_\alpha) = \sum \beta_i \underline{\mu}_i$$

$$V(\underline{y}_\alpha) = \left(\sum \frac{\beta_i^2 N_1}{N_i} \Sigma_i \right) \delta_{\alpha\beta}$$

Distribution of sample correlation coefficient

⇒ Karl Pearson Correlation coeff in bivariate sample

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 . \sum (y_i - \bar{y})^2}} = \frac{a_{12}}{\sqrt{a_{11} . a_{22}}}$$

$A \quad p = 2$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$r = f(a_{11}, a_{22}, a_{12}) \equiv f(a_{11}, a_{22}, r\sqrt{a_{11} . a_{22}})$$

→ Wishart dist.

$$f(A) = \frac{|A|^{\frac{n-p-1}{2}} \exp\left(-\frac{1}{2} \operatorname{tr} \Sigma^{-1} A\right)}{2^{\frac{np}{2}} . \prod_{i=0}^{p-1} \Gamma\left(\frac{n-i}{2}\right) \pi^{\frac{p(p-1)}{4}} |\Sigma|^{n/2}}$$

$p = 2$

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \quad \underline{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$f(A) = f(a_{11}, a_{22}, a_{12})$$

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11} a_{22} - a_{12}^2$$

$$|\Sigma| = \sigma_1^2 \sigma_2^2 (1 - \rho^2) \qquad \rho = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

$$|\Sigma^{-1}| = \frac{1}{(1-\rho)^2} \begin{vmatrix} \frac{1}{\sigma_1^2} & \frac{-\rho}{\sigma_1 \sigma_2} \\ & \frac{1}{\sigma_2^2} \end{vmatrix} = \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)}$$

$$\Sigma^{-1}A = \frac{1}{(1-\rho^2)}\begin{pmatrix} 1/\sigma_1^2 & -\rho/\sigma_1\sigma_2 \\ -\rho/\sigma_1\sigma_2 & 1/\sigma_2^2 \end{pmatrix}\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$= \frac{1}{(1-\rho^2)}\begin{pmatrix} \frac{a_{11}}{\sigma_1^2} - \frac{a_{21}\rho}{\sigma_1\sigma_2} & \frac{a_{21}}{\sigma_1^2} - \frac{\rho a_{22}}{\sigma_1\sigma_2} \\ -\frac{a_{11}\rho}{\sigma_1\sigma_2} + \frac{a_{21}}{\sigma_2^2} & -\frac{a_{12}\rho}{\sigma_1\sigma_2} + \frac{a_{22}}{\sigma_2^2} \end{pmatrix}$$

$$tr(\Sigma^{-1}A) = \frac{1}{(1-\rho^2)}\left[\frac{a_{11}}{\sigma_1^2} + \frac{a_{22}}{\sigma_2^2} - \frac{2a_{12}\rho}{\sigma_1\sigma_2}\right] \equiv Q$$

$$f(A) = \frac{(a_{11}a_{22} - a_{12}^2)^{\frac{n-3}{2}} \exp(-Q/2)}{2^n\sqrt{\pi}\,\Gamma(n/2).\Gamma((n-1)/2)\,|\Sigma|^{n/2}}$$

$$a_{12} = r\sqrt{a_{11}a_{22}} \qquad U = a_{11} \qquad |J| = \sqrt{a_{11}a_{22}}$$

$$da_{12} = \sqrt{a_{11}a_{22}}\,dr \qquad V = a_{22}$$

$$w = r\sqrt{a_{11}a_{22}}$$

$$f(A) = \overbrace{\frac{(a_{11}a_{22} - r^2a_{11}a_{22})^{\frac{n-3}{2}}}{2^n\sqrt{\pi}\,\Gamma(n/2).\Gamma((n-1)/2)\,|\Sigma|^{n/2}}}^{L^*} \cdot |J| . \exp\left[\frac{-1}{2(1-\rho^2)}\left\{\frac{a_{11}}{\sigma_1^2} + \frac{a_{22}}{\sigma_2^2} - \frac{2\rho r\sqrt{a_{11}a_{22}}}{\sigma_1\sigma_2}\right\}\right]$$

$$f(A) = f(a_{11}, a_{22}, r)$$

$$= L^* . \exp\left[\frac{-1}{2(1-\rho^2)}\frac{a_{11}}{\sigma_1^2}\right] . \exp\left[\frac{\rho r}{\sigma_1\sigma_2}.\frac{\sqrt{a_{11}a_{22}}}{(1-\rho^2)}\right]\sqrt{a_{11}a_{22}}$$

$$= \frac{a_{11}^{n/2-1}\,a_{22}^{n/2-1}(1-r^2)^{\frac{n-3}{2}}}{2^n|\Sigma|^{n/2}\sqrt{\pi}\,\Gamma(\frac{n}{2}).\Gamma(\frac{n-1}{2})} . \exp\left[\frac{-a_{11}}{2\sigma_1^2(1-\rho^2)}\right]\exp\left[\frac{-a_{22}}{2\sigma_2^2(1-\rho^2)}\right] . \sum_{\alpha=0}^{\infty}\left[\left\{a_{11}^{\frac{n+\alpha}{2}-1}\right.\right.$$

$$\left.\left.\exp\left(\frac{-a_{11}}{2(1-\rho^2)\sigma_1^2}\right)\right\}\left\{a_{22}^{\frac{n+\alpha}{2}-1}\exp\left(\frac{-a_{22}}{2(1-\rho^2)\sigma_2^2}\right)\right\}\left\{\left(\frac{\rho r}{\sigma_1\sigma_2(1-\rho^2)}\right)^{\alpha}\frac{1}{\alpha!}\right\}\right]$$

$$f(r) = \int_0^\infty\int_0^\infty f(a_{11}, a_{22}, r)\,da_{11}\,da_{22}$$

$$\# \int_0^\infty a_{11}^{\frac{n+\alpha}{2}-1}\exp\left(\frac{-a_{11}}{2(1-\rho^2)\sigma_1^2}\right)da_{11} = \Gamma\left(\frac{n+\alpha}{2}\right)\left[2\sigma_1^2(1-\rho^2)\right]^{\frac{n+\alpha}{2}}$$

$$\int_0^\infty a_{22}^{\frac{n+\alpha}{2}-1} \exp\left(-\frac{a_{22}}{2(1-\rho^2)\sigma_2^2}\right) da_{22} = \Gamma\left(\frac{n+\alpha}{2}\right)\left[2\sigma_2^2(1-\rho^2)\right]^{\frac{n+\alpha}{2}}$$

$$f(r) = \frac{(1-r^2)^{\frac{n-3}{2}}}{2^{n/2}\cdot\Gamma\left(\frac{n}{2}\right)\cdot\Gamma\left(\frac{n-1}{2}\right)\sqrt{\pi}\,\sigma_1^2\sigma_2^2(1-\rho^2)} \cdot \sum_{\alpha=0}^{\infty}\left[\frac{(\rho r)^\alpha}{\sigma_1^\alpha\sigma_2^\alpha(1-\rho^2)^\alpha\alpha!}\right] \rightarrow$$

$$\rightarrow \left[\left(\Gamma\left(\frac{n+\alpha}{2}\right)\right)^2 2^{n+\alpha}\cdot\sigma_1^{n+\alpha}\cdot\sigma_2^{n+\alpha}(1-\rho^2)^{n+\alpha}\right]$$

$$= \frac{(1-r^2)^{\frac{n-3}{2}}\,2^n\sigma_1^n\sigma_2^n(1-\rho^2)^n}{2^n\,\Gamma\left(\frac{n}{2}\right).\Gamma\left(\frac{n-1}{2}\right).\sqrt{\pi}\left(\sigma_1^2\sigma_2^2(1-\rho^2)\right)^{n/2}} \times \sum\left[\frac{\left(\Gamma\left(\frac{n+\alpha}{2}\right)\right)^2(\rho r)^\alpha 2^\alpha\sigma_1^\alpha\sigma_2^\alpha\cdot(1-\rho^2)^\alpha}{\sigma_1^\alpha\sigma_2^\alpha(1-\rho^2)^\alpha\alpha!}\right]$$

$$= \frac{(1-r^2)^{\frac{n-3}{2}}\,2^n\cdot\sigma_1^n\sigma_2^n(1-\rho^2)^n}{2^n\,\Gamma\left(\frac{n}{2}\right).\Gamma\left(\frac{n-1}{2}\right).\sqrt{\pi}\,\sigma_1^n\sigma_2^n(1-\rho^2)^{n/2}}\left[\sum\frac{(2\rho r)^\alpha}{\alpha!}\left(\Gamma\left(\frac{n+\alpha}{2}\right)\right)^2\right]$$

$$= \frac{(1-r^2)^{\frac{n-3}{2}}(1-\rho^2)^{n/2}}{\Gamma(n/2).\Gamma\left(\frac{n-1}{2}\right).\sqrt{\pi}}\sum_{\alpha=0}^{\infty}\left[\frac{(2\rho r)^\alpha}{\alpha!}\cdot\left(\Gamma\left(\frac{n+\alpha}{2}\right)\right)^2\right]$$

Duplication formula: $\Gamma(m).\Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(2m)}{2^{2m-1}}$

Replace $m \rightarrow \frac{n-1}{2}$ $\quad \Gamma\left(\frac{n-1}{2}\right).\Gamma\left(\frac{n-1}{2}+\frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(n-1)}{2^{n-2}}$

$$f(r) = \frac{(1-r^2)^{\frac{n-3}{2}}(1-\rho^2)^{n/2}\,2^{n-2}}{\sqrt{\pi}\sqrt{\pi}\,\Gamma(n-1)}\sum_{\alpha=0}^{\infty}\frac{(2\rho r)^\alpha}{\alpha!}\left(\Gamma\left(\frac{n+\alpha}{2}\right)\right)^2$$

$$f(r) = \frac{2^{n-2}(1-r^2)^{\frac{n}{2}-3}\,2^{n-2}}{\pi\,(n-2)!}\sum_{\alpha=0}^{\infty}\frac{(2\rho r)^\alpha}{\alpha!}\left(\Gamma\left(\frac{n+\alpha}{2}\right)\right)^2$$

where $n = N-1$

- The only estimator of ρ which can estimate ρ unbiasedly is $\sin^{-1} r$

$\mathbb{E}[\sin^{-1} r] = \sin^{-1}\rho \qquad \mathbb{E}(r) \neq \rho$

$$\mathbb{E}(r) \approx \rho - \frac{\rho(1-\rho^2)}{2n} + O\left(\frac{1}{n^2}\right)$$

$$V(r) \approx \frac{(1-\rho^2)^2}{n}\left(1 + \frac{11\rho^2}{2N}\right) + O\left(\frac{1}{n^3}\right)$$

Transform ρ to $z = \frac{1}{2}\log_e\left(\frac{1+\rho}{1-\rho}\right)$

Convergence to normality is more rapid on z transformation.